

Dowker Persistence and Hitting Times in Markov Chains

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Abstract. We develop a framework for studying persistent homology on probabilistically weighted directed networks. Directed networks are often asymmetric, causing standard topological constructions to be insufficient for capturing their structure. We extend asymmetrically weighted Dowker persistence to probabilistic directed networks using a negative logarithm thereby converting multiplicative path probabilities into additive weights suitable for persistent homology while preserving their ordering. We focus on directed cycle networks and investigate how their one-dimensional Dowker homology changes when a reverse edge is introduced. We then establish a connection between the resulting persistent homology and the hitting times of the associated Markov chain. Specifically, we prove that the death time of the one dimensional Dowker homology is determined by the pair of vertices minimizing the maximum of their mutual hitting times.

Key words. Dowker persistence, weighted cycles, Markov chains, directed graphs.

AMS subject classifications. 55N31, 05C20, 05C90, 60J22

1. Introduction. Many systems can be modeled as directed networks where relationships between states are both asymmetric and probabilistic [21, 17]. Examples include communication systems, sensor networks, climate dynamics [10], and stochastic processes modeled by Markov chains [16]. Understanding the global structure of such systems requires tools capable of capturing the macro scale trends that emerge from these directed interactions. Topological Data Analysis (TDA) provides a powerful mathematical framework for this purpose by evaluating the underlying shape of data through the construction of simplicial complexes [3, 8]. While standard methods in TDA, such as the Vietoris-Rips complex, are blind to asymmetry, recent developments in Dowker persistence have provided a framework for studying directed relationships [7, 6].

This work builds directly upon the framework of asymmetrically weighted Dowker persistence introduced by Timofeyev et al. [21], who provided a characterization of one-dimensional Dowker homology for directed cycle graphs representing dynamical trajectories. While their work established how topological signatures reflect the flow and spatial organization of binned dynamical data, it primarily addressed deterministic weighted networks. In that paper, the authors focus on directed cycles, whereas here we examine how the one dimensional Dowker homology changes when a reverse edge is introduced.

In this paper we extend the framework of Dowker persistence to probabilistic directed networks by replacing deterministic edge weights with transition probabilities. A central challenge in this setting which we address is that transition probabilities combine multiplicatively along paths, whereas the simplicial filtration used in persistent homology requires additive

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edge weights. This discrepancy is addressed with a negative logarithm transformation, which converts transition probabilities into additive weights while preserving their relative ordering. Under this transformation, highly probable transitions correspond to small weights, and less probable transitions correspond to larger weights. A similar idea can be found for a more specific biomolecular setting in [5]. After the transform, the resulting Dowker complex filtration provides a topological representation of the probabilistic connectivity of the original system.

This allows the network to be viewed naturally as a Markov chain on a coarse grained state space. A Markov chain is a collection of states and transition probabilities between them [16]. The identifying feature of a Markov chain is that the next state depends only on the current state and not on any of the previous states, making Markov chains a memoryless process. Markov chains have been studied since the early 20th century and have found applications in statistics with Markov Chain Monte Carlo methods [11], information theory [19], climate science [20], and symbolic dynamics [14]. The hitting time of a Markov chain refers to the expected time it takes for the Markov chain to reach a target state from a starting state [16].

Our main contribution is to establish a direct relationship between persistent homology and expected hitting times of the underlying Markov chain. For directed cycle networks, we prove that the death time of the one dimensional homology is determined by the pair of vertices that minimizes the maximum mutual hitting time, when two vertices can jointly acquire all information in the network as quickly as possible.

In section 2, we go over background information on persistent homology, followed by specific background of Dowker persistence on asymmetric networks, in the style of [6]. Then, we present the logarithm transform for probabilistic persistent homology, and characterize the 1D barcode in a way analogous to [21]. After that, we share some results on what happens to the homology after adding an edge traveling in the reverse direction in section 3. In section 4, we present some background on hitting times in Markov chains and briefly discuss our algorithm for computing the hitting time matrix. Then, the main result connecting persistent homology to hitting times is in section 5.

2. Persistent Homology. Persistent homology is a central tool in topological data analysis that quantifies the topological structure of data by tracking the evolution of features such as connected components, cycles, and higher-dimensional voids across a filtration of simplicial complexes. Building on foundational ideas in algebraic topology [12] and computational topology [8], the theory was formalized using topological persistence, a framework that tracks when features appear and disappear as a scale parameter changes [9]. Efficient representations of simplicial complexes, such as the simplex tree [2], enable practical computation, while extensions to directed and asymmetric settings [6] broaden its applicability to networked and dynamical data. Persistent homology has proven particularly powerful in analyzing complex systems and time series, including applications to dynamical systems and climate phenomena [17, 10], as well as identifying recurring patterns such as cycles in data [1]. Its robustness to noise and ability to extract qualitative geometric information make it a framework used across many scientific domains [3]. To formalize this framework, we will briefly introduce the following foundational concepts.

A simplex σ is the basic geometric building block of a complex. A k -dimensional simplex is defined as the convex hull of $k+1$ affinely independent points $\{v_0, v_1, \dots, v_k\} \subset \mathbb{R}^d$.

$$(2.1) \quad \sigma = \text{conv}\{v_0, \dots, v_k\} = \left\{ \sum_{i=0}^k \lambda_i v_i \mid \sum_{i=0}^k \lambda_i = 1, \lambda_i \geq 0 \right\},$$

where $\dim(\sigma) = k$. The points are affinely independent if the vectors $\{v_i - v_0\}_{i=1}^k$ are linearly independent. Low dimensional examples include:

$k = 0$	(0-simplex)	$\sigma = [v_0]$	a vertex
$k = 1$	(1-simplex)	$\sigma = [v_0, v_1]$	an edge
$k = 2$	(2-simplex)	$\sigma = [v_0, v_1, v_2]$	a triangle
$k = 3$	(3-simplex)	$\sigma = [v_0, v_1, v_2, v_3]$	a tetrahedron

A simplicial complex K is a finite collection of simplices in $\mathbb{R}^d$ that satisfies two conditions:

1. If $\sigma \in K$ and $\tau \subset \sigma$, then $\tau \in K$. We say τ is a face of σ
2. For any $\sigma_1, \sigma_2 \in K$, either $\sigma_1 \cap \sigma_2 = \emptyset$ or $\sigma_1 \cap \sigma_2$ is a face of both σ_1 and σ_2 .

The underlying space of K , denoted by $|K|$, is the union of all simplices contained in K ,

$$(2.2) \quad |K| = \bigcup_{\sigma \in K} \sigma.$$

This structure allows us to represent complex topological spaces using discrete combinatorial data where each point $x \in |K|$ belongs to the interior of exactly one simplex $\sigma \in K$.

A filtration represents the growth history of a simplicial complex K , where every simplex σ is given an order of entry, usually with a positive real value weight. It is formally defined as a nested sequence of subcomplexes indexed by a scale parameter δ :

$$(2.3) \quad \emptyset = K_{\delta_0} \subseteq K_{\delta_1} \subseteq \dots \subseteq K_{\delta_n} = K,$$

where the inclusion is controlled by the ordering of the parameter:

$$(2.4) \quad \delta_i \leq \delta_j \implies K_{\delta_i} \subseteq K_{\delta_j}.$$

As the scale parameter δ increases, additional simplices are added to the complex, causing the underlying topological space $|K_\delta|$ to evolve and revealing structural features across multiple scales.

Homology is the algebraic formalism used to quantify structural “holes” in a topological space. For a simplicial complex K , let $C_k(K)$ denote the k -th chain group, a vector space over a field $\mathbb{F}$ (commonly $\mathbb{Z}_2$), generated by the oriented k -simplices of K as a basis:

$$(2.5) \quad C_k(K) = \left\{ \sum_i a_i \sigma_i \mid a_i \in \mathbb{F}, \sigma_i \in K_k \right\}.$$

The relationships between adjacent dimensions are governed by the boundary operator

$$(2.6) \quad \partial_k : C_k(K) \rightarrow C_{k-1}(K),$$

which is defined on a k -simplex as the alternating sum of its faces

$$(2.7) \quad \partial_k[v_0, v_1, \dots, v_k] = \sum_{i=0}^k (-1)^i [v_0, v_1, \dots, \widehat{v_i}, \dots, v_k],$$

where $\widehat{v_i}$ denotes the omission of the i -th vertex. A fundamental property of the boundary operator is that the boundary of a boundary is trivial: $\partial_{k-1} \circ \partial_k = 0$. This relation defines two important subgroups: k -cycles, $Z_k(K) = \ker(\partial_k)$, which identify chains with zero boundary and k -boundaries, $B_k(K) = \text{im}(\partial_{k+1})$, which identify cycles that are themselves boundaries of $(k+1)$ dimensional structures. The k th homology group $H_k(K)$ is defined as the quotient space of cycles modulo boundaries:

$$(2.8) \quad H_k(K) = \frac{Z_k(K)}{B_k(K)}.$$

Elements of $H_k(K)$ are homology classes representing k -dimensional cycles that do not bound a higher-dimensional structure, thereby capturing the structural cavities of the complex. The rank of this group is the k -th Betti number,

$$(2.9) \quad \beta_k = \dim H_k(K).$$

This quantity counts the number of independent k -dimensional holes in the complex: β_0 counts connected components, β_1 counts one-dimensional loops or cycles, and β_2 counts two-dimensional voids or cavities.

Persistent homology studies how the homology of a space changes as a simplicial complex grows through a filtration. The key idea is that homology is functorial: if $f : X \rightarrow Y$ is a simplicial map, then for each dimension k there is an induced map on homology,

$$(2.10) \quad H_k(f) : H_k(X) \rightarrow H_k(Y).$$

A filtration is a nested sequence of simplicial complexes

$$(2.11) \quad \emptyset = X_0 \subseteq X_1 \subseteq \dots \subseteq X_n,$$

where each inclusion $X_i \hookrightarrow X_{i+1}$ induces a homomorphism

$$(2.12) \quad \mu_{i,j} : H_k(X_i) \rightarrow H_k(X_j), \quad i \leq j.$$

For a fixed dimension k , the collection of homology groups together with the induced maps,

$$(2.13) \quad \{H_k(X_i), \mu_{i,j}\}_{i \leq j},$$

is called the *persistence module*. It describes how connected components, cycles, and higher-dimensional voids evolve as the filtration parameter increases.

A homology class is said to be *born* at filtration value δ_b if it first appears in the filtration at parameter δ_b , and it *dies* at filtration value $\delta_d > \delta_b$ when it either becomes the boundary of a higher-dimensional simplex or merges with an older homology class. The ordered pair (δ_b, δ_d) is

called the *persistence pair*, and the quantity $\delta_d - \delta_b$ is the *persistence* of the feature. Persistent homology is commonly summarized by a *persistence barcode*, in which each homology class is represented by an interval $[\delta_b, \delta_d)$.

Features with long persistence are typically interpreted as significant topological structures, whereas short-lived features are often regarded as topological noise. The definitions presented in this section follow standard treatments of algebraic topology and persistent homology. Readers interested in additional details can consult Hatcher [12] for algebraic topology, Edelsbrunner and Harer [8] for computational topology and persistent homology, and Edelsbrunner, Letscher, and Zomorodian [9] for topological persistence.

2.1. Asymmetric Networks. Persistent homology can be extended to asymmetric (directed) networks, where the lack of symmetry requires constructions beyond standard simplicial complexes. One commonly employed method is given by Dowker complexes, originating from the work of Dowker [7], which associate dual simplicial complexes to a relation and are perfectly suited to directed data. Formally, let

$$R : X \times Y \rightarrow \{0, 1\}$$

be a relation. The *Dowker source complex* is

$$D^{\text{so}}(R) = \{\sigma \subseteq X \mid \exists y \in Y, \forall x \in \sigma, R(x, y) = 1\}$$

and the *Dowker sink complex* is

$$D^{\text{si}}(R) = \{\tau \subseteq Y \mid \exists x \in X, \forall y \in \tau, R(x, y) = 1\}.$$

By Dowker's Theorem

$$D_{\text{so}}(R) \simeq D_{\text{si}}(R).$$

Given a weighted directed network

$$G = (X, \omega) \quad \omega : X \times X \rightarrow [0, \infty),$$

let

$$P(G) = (X, d)$$

denote its path completion, where

$$d(x, y) = \inf_{p: x \rightsquigarrow y} \omega(p)$$

is the shortest path distance. The function d is a Lawvere metric, allowing asymmetry and infinite values. For $\delta \geq 0$, let

$$R_\delta(x, y) = \mathbf{1}_{\{d(x, y) \leq \delta\}}.$$

This induces the filtration

$$\left\{ D^{\text{so}}(R_\delta(P(G))) \right\}_{\delta \in \mathbb{R}_{\geq 0}}$$

with

$$\delta_1 \leq \delta_2 \implies D^{\text{so}}(R_{\delta_1}) \subseteq D^{\text{so}}(R_{\delta_2}).$$

Persistent homology tracks the birth and death of homological features as δ varies, capturing directed cycles and higher order structures. Thus, Dowker persistence provides a natural framework for studying the topology of weighted directed networks.

2.2. Probabilistic Approach. Considering a probabilistic approach to persistent homology, particularly in asymmetric networks, extends the topological framework and allows it to directly encode the likelihood and dynamics of information flow. Weighting directed edges using transition probabilities essentially treats the network as a Markov chain, allowing for the use of persistent homology to study Markov chains. This has been done in slightly different ways before to study transportation networks [4] and convection in networks [15]. This framework provides a rigorous topological signature for decentralized communication. This probabilistic framing is particularly valuable for identifying critical milestones like the death of the 1-cycle, which, as we will show, characterizes the point of two-point domination or collective consensus between specific sinks/sources.

A negative log transformation is necessary for the Dowker filtration because it bridges the gap between multiplicative probabilities and the additive distance structure required for persistent homology. In a network modeled as a Markov chain, the probability of traversing a path $\gamma = (e_1, e_2, \dots, e_k)$ is given by

$$(2.14) \quad P(\gamma) = \prod_{i=1}^k p_i,$$

where p_i denotes the transition probability associated with edge e_i . However, the Dowker persistence framework is built on a weighted shortest path metric, where distances are defined additively [21]. To address this, transition probabilities are transformed into edge weights via

$$(2.15) \quad w_i = -\log(p_i).$$

Under this transformation, the sum of the edge weights is the negative log probability of the path:

$$(2.16) \quad -\log P(\gamma) = -\log \left(\prod_{i=1}^k p_i \right) = \sum_{i=1}^k -\log(p_i) = \sum_{i=1}^k w_i,$$

which is compatible with additive path metrics. This mapping sends high probability transitions ($p_i \approx 1$) to small weights ($w_i \approx 0$), and low probability transitions ($p_i \approx 0$) to large weights ($w_i \gg 1$). As δ increases, the filtration progressively includes paths of decreasing probability, and the birth and death times of homological features correspond to thresholds in probabilistic connectivity.

2.3. Persistence Conditions Under Probabilistic Weighting. The logarithmic transformation allows existing results from asymmetrically weighted Dowker persistence to be translated into the probabilistic setting. In particular, the criterion for nontrivial one-dimensional persistence on directed cycle graphs established in [21] can be reformulated in terms of transition probabilities. The translated result states that Markov chains from consistently oriented cycle graphs only have a 1st dimensional persistence class if there is not a highly improbable edge. Here, highly improbable would mean an edge with a lower probability than the product of all other edges. Thus, if it is more likely to traverse the cyclic Markov chain from state i to $i - 1$ than it is from $i - 1$ to i , no 1D bar will ever be born for that Markov chain.

Proposition 2.1 (Probabilistic Cycle Criterion, adapted from [21]). *Let G be a directed cycle graph with edge weights*

$$w_i = -\log(p_i),$$

where p_i denotes the transition probability along edge i . Then G has nontrivial 1 dimensional homology if and only if for every edge i

$$w_i < \sum_{j \neq i} w_j$$

Equivalently,

$$p_i > \prod_{j \neq i} p_j$$

Proof. Let G be a cycle digraph with n nodes. For each directed edge from v_i to v_{i+1} , define the edge weight

$$(2.17) \quad w(v_i, v_{i+1}) = -\log P(v_i \rightarrow v_{i+1}),$$

where $P(v_i \rightarrow v_{i+1})$ is the transition probability of moving from v_i to v_{i+1} .

Since $P_i \in (0, 1]$, we have $w_i \geq 0$ for all i , so the inequality

$$(2.18) \quad p_i > \prod_{j \neq i} p_j$$

holds if

$$(2.19) \quad w_i < \sum_{j \neq i} w_j \quad \text{for all } i.$$

w_i must be strictly less than $\sum_{j \neq i} w_j$, because when they are equal, edge i enters the filtration at the same time as the complementary path. Thus the loop is completed at the same time as it is filled in, meaning no 1-dimensional feature ever forms.

Going from weighted edges back to edge probabilities,

$$(2.20) \quad e^{-\log(P_i)} < \prod_{j \neq i} e^{-\log(P_j)}.$$

Using the identity $e^{-\log(x)} = x^{-1}$, this simplifies to

$$(2.21) \quad \frac{1}{P_i} < \prod_{j \neq i} \frac{1}{P_j}.$$

Thus, the condition (2.19) holds if and only if

$$(2.22) \quad P_i > \prod_{j \neq i} P_j \quad \text{for all } i.$$

If (2.22) fails for some i , then the complementary path enters the filtration before edge i , causing the loop to be filled in before its boundary is completed. No 1-dimensional feature forms. Conversely, if (2.22) holds for every i , then each edge appears before its complementary path, so the boundary of the cycle is completed strictly earlier than the filling, producing a nontrivial persistent 1-cycle.

Thus the cycle has nontrivial 1-dimensional homology if and only if its maximum edge weight is strictly smaller than the sum of the remaining edge weights. When translated back to probability, this means that there is a nontrivial 1 dimensional homology if and only if the least probable edge is more likely than the product of all of the other edges. ■

Definition 2.2 (Dowker Death Time). *Let W denote the distance value matrix. The 1-cycle dies when the cycle is fully filled in, which requires the edge with the largest filtration value in either direction for some pair (a, b) to appear. Thus, the death time is the minimal d such that*

$$(2.23) \quad d \geq \max(W_{a,b}, W_{b,a})$$

for some pair (a, b) , taking the minimal such d over all pairs, we obtain

$$(2.24) \quad \text{Death Time} = \min_{a,b} \max(W_{a,b}, W_{b,a}) \quad [21]$$

3. Perturbations by Bidirectional Edges. Most existing work on the analysis of networks with persistent homology, even using asymmetric homology theories like in [6, 21], still only allows for flow in one direction between two adjacent nodes, i.e. if there exists an edge $a \rightarrow b$, then there is no edge $b \rightarrow a$. However, this perspective doesn't account for relationships that arise in systems where transitions occur in both directions with potentially different likelihoods. To capture this added structure, we extend the framework to incorporate a bidirectional edge, allowing a pair of nodes to be connected by asymmetric transition probabilities in each direction. This is considered less explicitly in [18] using a different filtration construction. This enables a more nuanced representation of connectivity and opens the door to identifying topological features that reflect directional imbalances and feedback within the network, revealing patterns that are not accessible under exclusively unidirectional models.

We will study a simple example: a four node cycle with a bidirectional edge. By introducing one edge whose direction opposes the dominant cyclic flow, we can explicitly track how

this perturbation alters the filtration and the resulting homology. In particular, we examine how the weight of the reverse transition affects the birth and death of the 1-cycle. Notably, the additional edge can only cause the cycle to be born earlier, die earlier, or fail to appear altogether. It can never delay either event, since adding simplices to a filtration can only decrease (never increase) the time at which a boundary becomes complete.

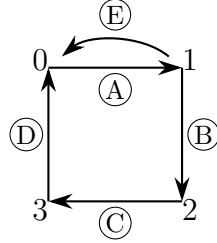


Figure 3.1. 4-cycle with one reverse edge.

We begin with the standard directed four-cycle which contains no diagonals. The introduction of a single reverse edge, shown in Figure 3.1, creates additional 2-simplices in the filtration: depending on the direction of the reverse transition, the filtration may generate the diagonal (and triangular 2-simplex) Δ_{BE} , while Δ_{BCE} denotes the 2-simplex completing the full loop. These simplices, and the other possible triangles, do not belong to the underlying graph; rather, they arise only through the filtration process once the reverse edge is included.

Theorem 3.1. *The inclusion of an edge E such that $P(C), P(D), P(\Delta_{BE}) > P(\Delta_{CD}) > P(A)$ induces an earlier birth of a 1-dimensional homology class.*

Proof. Assume the following:

$$P(C), P(D), P(\Delta_{BE}) > P(\Delta_{CD}) > P(A)$$

This means that without the edge E , the cycle is born at $P(A)$, since A is the least probable of the original four edges. However, adding E such that Δ_{BE} enters the filtration before A and Δ_{CD} means that $\{C, D, \Delta_{BE}\}$ is a one-cycle born before $\{A, B, C, D\}$. An example of this can be seen in Figure 3.2

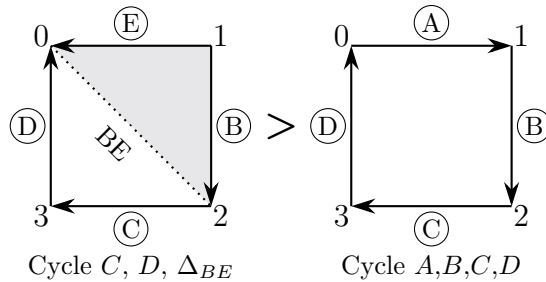


Figure 3.2. Half loop versus full loop. The edge Δ_{BE} advances the birth time of the loop.

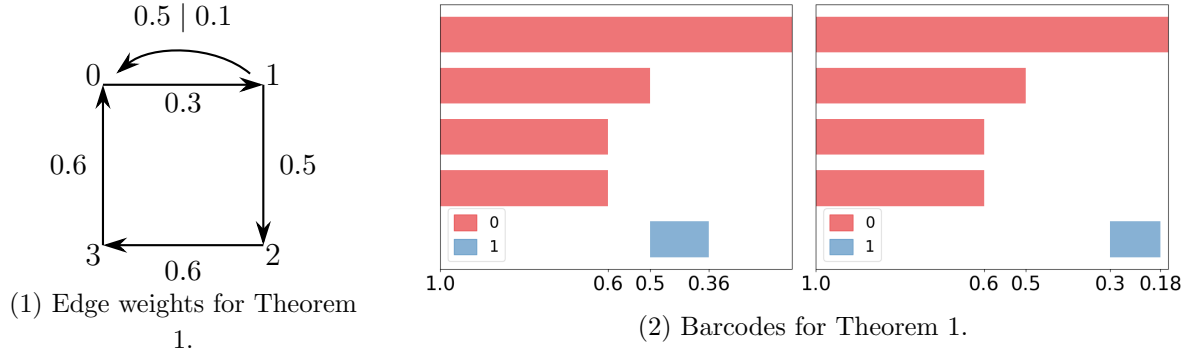


Figure 3.3. Graph and corresponding barcodes for Theorem 1.

In Figure 3.3.1 the weights of A and B are 0.3 and 0.5, respectively. The weight 0.1 for E corresponds to the right barcode shown in Figure 3.3.2 and illustrates the case in which the reverse edge E does not affect the homology. The inclusion of the reverse edge E at a weight of .5 causes the one cycle $\{C, D, \Delta_{BE}\}$ to be born at

$$\min\{0.6, 0.6, 0.5\} = 0.5$$

This occurs strictly earlier than the birth of the full cycle $\{A, B, C, D\}$, which is born at the least probable boundary edge,

$$\min\{0.3, 0.5, 0.6, 0.6\} = 0.3$$

which is given in Figure 3.3.2. In this example, the reverse edge E has no effect until $P(E) > 0.3$.

Theorem 3.2. *The inclusion of an edge E such that $P(\Delta_{BE}), P(\Delta_{CD}) > P(\Delta_{AB})$ can also decrease the death time of the corresponding 1-dimensional homology class.*

Proof. A. Suppose the following probabilities are given:

$$P(\Delta_{CD}) > P(\Delta_{BE}) > P(\Delta_{AB})$$

$P(\Delta_{CD})$ enters the filtration earliest, therefore is included before $P(\Delta_{BE})$ and finally $P(\Delta_{AB})$. The boundary determined by $\{\Delta_{CD}, \Delta_{BE}\}$ is therefore completed strictly before $P(\Delta_{AB})$ appears, demonstrating that the inclusion of edge E decreases the death time. See Figure ?? for an example. ■

In Figure 3.5.1 the weights of A and B are 0.5 and 0.5, respectively. The weight 0.1 corresponds to the right barcode shown in Figure 3.5.2 and illustrates the case in which the reverse edge E does not affect the homology. The inclusion of the reverse edge E at a weight of .4 causes the one cycle $\{A, B, \Delta_{CD}\}$ to die sooner at

$$\min\{\Delta_{CD}, \Delta_{BE}\} = \min\{0.6 \times 0.7, 0.4, 0.5\} = 0.4$$

rather than later at

$$\min\{\Delta_{CD}, \Delta_{AB}\} = \min\{0.6 \times 0.7, 0.5 \times 0.5\} = 0.25$$

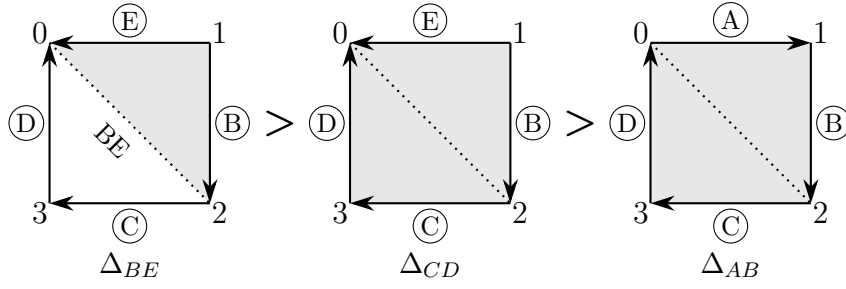


Figure 3.4. Illustration of $P(\Delta_{BE})$ entering the filtration first as $\{\Delta_{CD}, \Delta_{BE}\}$ complete the boundary first.

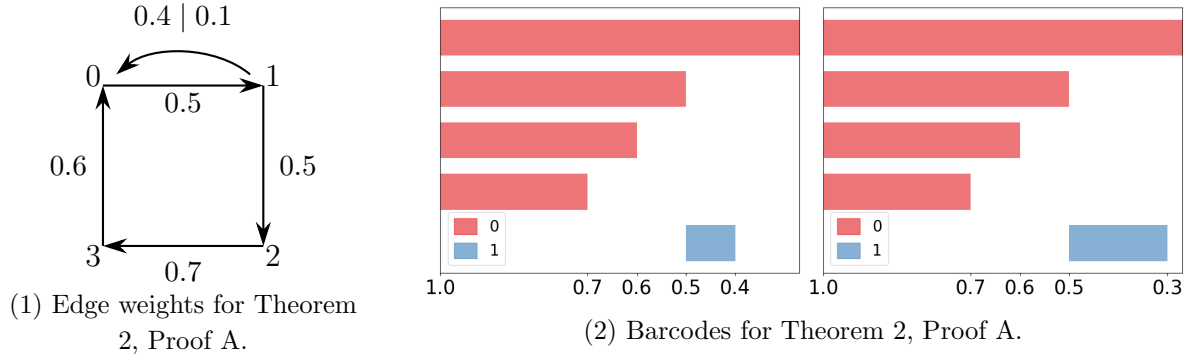


Figure 3.5. Graph and corresponding barcodes for Theorem 2A.

Although in this specific example, the two other triangles Δ_{BC} and Δ_{AD} are more probable which causes the loop to die at 0.3:

$$\min\{\Delta_{BC}, \Delta_{AD}\} = \min\{0.5 \times 0.7, 0.6 \times 0.5\} = 0.3$$

Proof. B. Consider the setup:

$$P(\Delta_{BE}) > P(\Delta_{CD}) > P(\Delta_{AB})$$

$P(\Delta_{BE})$ enters the filtration earliest, therefore is included before $P(\Delta_{CD})$ and finally $P(\Delta_{AB})$. The boundary determined by $\{\Delta_{CD}, \Delta_{BE}\}$ is therefore completed strictly before $P(\Delta_{AB})$ appears, demonstrating that the inclusion of edge E decreases the death time. This is shown in Figure 3.6. ■

In Figure 3.7.1 the weights of A and B are 0.5 and 0.5, respectively. The weight 0.1 corresponds to the barcode shown in the right of Figure 3.7.2 and illustrates the case in which the reverse edge E does not affect the homology. The inclusion of the reverse edge E causes the one cycle $\{C, D, \Delta_{BE}\}$ to die sooner at

$$\min\{\Delta_{BE}, \Delta_{CD}\} = \min\{0.5, 0.5, 0.6 \times 0.6\} = 0.36$$

rather than later at

$$\min\{\Delta_{BE}, \Delta_{AB}\} = \min\{0.5, 0.5 \times 0.5\} = 0.25$$

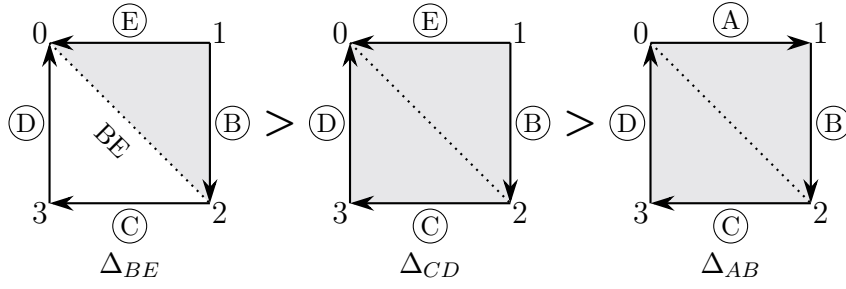


Figure 3.6. Illustration of $P(\Delta_{BE})$ entering the filtration first as $\{\Delta_{CD}, \Delta_{BE}\}$ complete the boundary first.

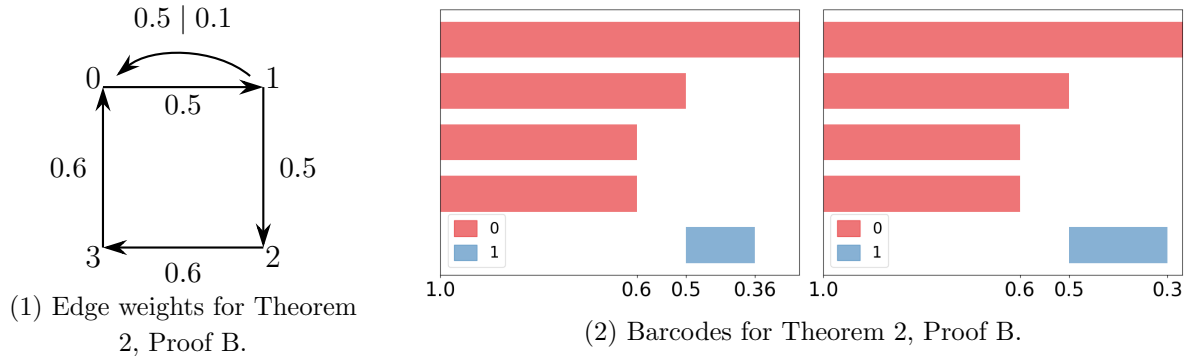


Figure 3.7. Graph and corresponding barcodes for Theorem 2B.

Although in this case, the two other triangles Δ_{BC} and Δ_{AD} are more probable, which causes the loop to die at 0.3.

$$\min\{\Delta_{BC}, \Delta_{AD}\} = \min\{0.5 \times 0.7, 0.6 \times 0.5\} = 0.3$$

Theorem 3.3. The inclusion of an edge E such that $P(BCE) > \min\{P(A), P(D)\}$ induces no 1-dimensional homology class to be born.

Proof. Suppose the following:

$$P(\Delta_{BE}), P(\Delta_{BC}) > P(D)$$

The 2-simplexes Δ_{BE} and Δ_{BC} enter the filtration before the final boundary edge D . As a result, the interior of the loop is filled prior to the completion of its boundary. This leads to no 1-dimensional homology class being born, as shown in Figure 3.8. ■

In Figure 3.9.1 the weights of A and B are 0.7 and 0.9, respectively. The weights 0.1 correspond to the right barcode shown in Figure 3.9.2 and represents the case in which the reverse edge E does not affect the homology. The inclusion of the reverse edge E at weight 0.8 causes the 2-simplex Δ_{BE} to be born at

$$\min(0.8, 0.9) = 0.8$$

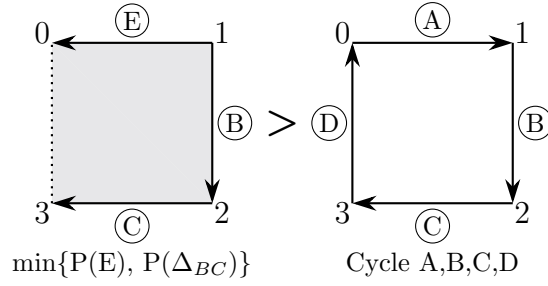


Figure 3.8. Illustration of loop, or 2-simplex, BCE being completely filled before the boundary is formed.

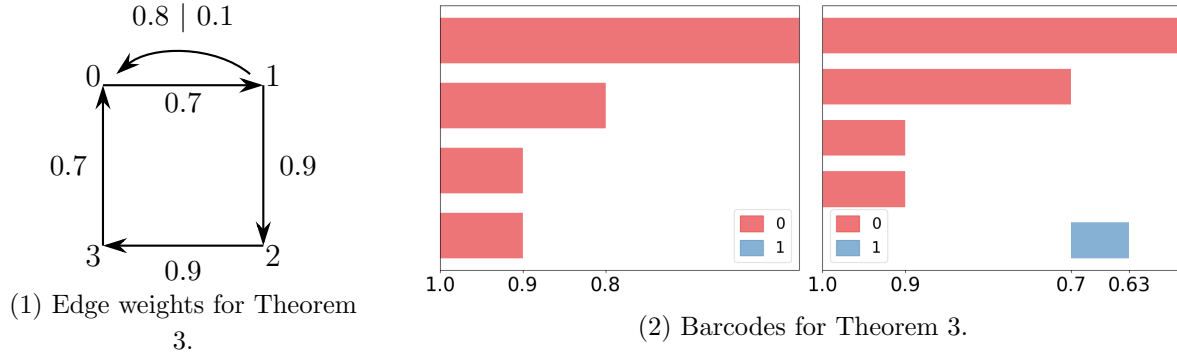


Figure 3.9. Graph and corresponding barcodes for Theorem 3.

and 2-simplex Δ_{BC} to be born at

$$0.8 \times 0.9 = .72,$$

but the cycle is not born until 0.7.

Since this occurs at a larger probability than the remaining boundary edge D , it causes the interior of the loop $\{A, B, C, D\}$ to be completed, or filled, as its boundary is completed. Consequently, the 1-dimensional homology class is never born. In this example, the reverse edge E has no effect until $P(E) > 0.77$.

4. Hitting Time. In the study of dynamical systems and decentralized networks, Markov chains provide the mathematical foundation for modeling state transitions and information flow. While transition probabilities describe the instantaneous behavior of the chain itself, many structural properties are not revealed by the probabilities themselves, but by the *expected hitting time* required to propagate between states. This is especially useful in decentralized networks where hitting times naturally quantify how long it takes, on average, for the Markov chain to reach a given state [16]. This allows us to use transitions as a way to analyze information flow. Here, we detail a connection between the hitting times of a cyclic Markov chain and its Dowker persistence.

In decentralized networks, a complete global map of the system is typically unavailable. Instead, one can perform local queries: for any node v , all nodes connected to it can be identified, at some cost. In this context, a natural question is how long it takes to reach a given target node when navigating through the network from a starting point. This is

measured by the *hitting time*. Let $(X_n)_{n \geq 0}$ be a Markov chain with state space S , and let $A \subseteq S$. The *hitting time* of A is defined as

$$(4.1) \quad T_A = \inf\{n \geq 0 : X_n \in A\}.$$

That is, T_A is the first time the Markov chain enters the set A . If the chain never reaches A , we set $T_A = \infty$. For a starting state $i \in S$, the *expected hitting time* of A is defined as $\mathbb{E}_i[T_A]$ where $\mathbb{E}_i$ denotes expectation conditioned on $X_0 = i$. This is the expected number of steps required for the chain, starting from state i , to reach the set A [16].

4.1. Hitting Time Algorithm. The algorithm computes the hitting time matrix HT of a Markov chain with transition matrix P . For each target state t , it first identifies the set R of states that can reach t with positive probability by iteratively expanding the reachable set. It then constructs the linear system corresponding to the hitting time equations on R :

$$(4.2) \quad h_i = 1 + \sum_j P_{ij} h_j,$$

with boundary condition $h_t = 0$. Solving this system yields the expected number of steps required to reach t from every state in R , while states outside R are assigned hitting time ∞ . Repeating this procedure for each target state produces the full hitting time matrix HT , where the entry H_{ij} denotes the expected time for the Markov chain, beginning in state i , to reach state j . The full procedure can be found implemented in Python at the CACTIS repository [13].

5. Persistent Homology and its Relation to Hitting Times. Hitting times quantify how long it takes a Markov chain to reach a given state, while persistent homology records when topological features such as cycles appear and disappear in a filtration. Although these notions arise from different mathematical frameworks, they coincide in a particular way for strictly directed cycles. In such cycles, the transition probabilities determine both the expected time required to traverse each edge and the Dowker filtration values assigned to those edges. As a result, the hitting time matrix encodes the birth of the fundamental 1-cycle and identifies the pair of vertices whose connection causes its death. We formalize this relationship below.

Proposition 5.1. *Let D be the Dowker filtration associated with a consistently oriented cycle. Then the birth simplex of the unique generator of $H_1(D)$ is the directed edge having the largest one step hitting time. If $HT = (H(i, j))$ is the hitting time matrix, then the birth simplex is determined by the largest adjacent entry of HT , specifically the entry on the cyclic first superdiagonal for forward edges.*

Proof. Consider a directed cycle with states

$$0, 1, \dots, n-1$$

For each adjacent pair of vertices, there may exist a forward edge, a reverse edge, or both. If both directed edges are present, we restrict our attention to the edge with the larger transition probability and discard the other. This produces a directed cycle with exactly one non self

outgoing edge from each vertex. This simplification is sufficient for identifying the birth simplex in the Dowker filtration. The Dowker filtration value of an edge is given by

$$(5.1) \quad w(i, i') = -\ln P(i, i')$$

and the birth simplex is the edge having the largest filtration value. Since $-\ln(x)$ is strictly decreasing on $(0, 1]$, the ordering of the filtration values is completely determined by the ordering of the transition probabilities. Consequently, selecting a single directed edge between each adjacent pair preserves the edge responsible for the birth of the unique generator of $H_1(D)$ if the edge selected is the more probable of the two.

Suppose vertex i has a directed transition to an adjacent vertex i' with probability $P(i, i')$. Then

$$(5.2) \quad H(i, i') = 1 + P(i, i)H(i, i') = 1 + (1 - P(i, i'))H(i, i').$$

Rearranging gives

$$(5.3) \quad P(i, i')H(i, i') = 1,$$

and hence

$$(5.4) \quad H(i, i') = \frac{1}{P(i, i')}.$$

Taking logarithms,

$$(5.5) \quad \ln(H(i, i')) = \ln\left(\frac{1}{P(i, i')}\right) = -\ln P(i, i') = w(i, i').$$

Thus, the logarithm of the one step hitting time coincides exactly with the Dowker edge weight. Since the logarithm is strictly increasing,

$$(5.6) \quad \arg \max_{(i, i')} H(i, i') = \arg \max_{(i, i')} w(i, i')$$

The unique generator of $H_1(D)$ is born when the final edge of the directed cycle enters the filtration. Consequently, the birth simplex is the edge having the largest filtration value, which is precisely the edge having the largest one step hitting time. Finally, if $HT = (H(i, j))$ denotes the hitting time matrix, then the one step hitting times occur on the cyclic first superdiagonal for forward edges and on the cyclic first subdiagonal for reverse edges. Therefore, the birth simplex is determined by the largest adjacent entry of HT . ■

Proposition 5.2. *For a directed cycle with edge weights derived from transition probabilities, the death simplex of the unique 1-cycle in the filtration is attained at the pair*

$$\arg \min_{i \neq j} \max(HT_{i,j}, HT_{j,i}).$$

That is, the pair (i, j) for which the larger of the two mutual hitting times is minimal, where HT denotes the hitting time matrix. Equivalently, the death simplex corresponds to the larger of the hitting times of the symmetric pair whose maximal mutual hitting time is smallest.

Proof. As shown in Proposition 5.1, if i' is the unique forward neighbor of i , then

$$(5.7) \quad H(i, i') = \frac{1}{P(i, i')},$$

and therefore

$$(5.8) \quad w(i, i') = -\ln P(i, i') = \ln H(i, i').$$

Hence the one step hitting times determine exactly the edge weights of the Dowker filtration. Now consider a path of any length,

$$(5.9) \quad i = v_0, v_1, \dots, v_k = j.$$

Since each vertex has exactly one outgoing edge, and they are doing in a consistent order, the path from i to j must traverse each intermediate vertex in order. Therefore,

$$(5.10) \quad H(i, j) = \sum_{r=0}^{k-1} H(v_r, v_{r+1}),$$

so every entry of the hitting time matrix is simply the sum of the one step hitting times along the unique directed path from i to j . Since the one step hitting times coincide with the Dowker edge weights, each hitting time coincides with the weighted length of the corresponding directed path in the filtration.

For any pair of vertices (i, j) , the quantities

$$H(i, j) \quad \text{and} \quad H(j, i)$$

are the weighted lengths of the two complementary directed paths obtained by cutting the cycle at i and j . Since these paths partition the cycle, their total weight is fixed. Thus, minimizing the larger of the two path lengths is equivalent to choosing the cut for which the longest complementary path is as short as possible. Therefore, the death simplex is attained at the pair

$$(5.11) \quad (i, j) = \arg \min_{i \neq j} \max\{H(i, j), H(j, i)\},$$

which is precisely the desired characterization. ■

5.1. Example: Hitting Times and the Death Simplex. We now illustrate the relationship between the Dowker death simplex and the hitting-time characterization established in Proposition 2. Consider the directed cycle graph shown in Figure 5.1. Applying the logarithmic transformation $w_i = -\log(p_i)$, produces a weighted Dowker filtration on the directed cycle. The resulting barcode contains a single persistent 1 dimensional class corresponding to the cycle.

To compare this filtration behavior with the associated Markov chain, we compute the hitting times between vertices, yielding the hitting time matrix,



Figure 5.1. H_1 feature persisting from 0.6 to 0.432.

$$(5.12) \quad H = \begin{matrix} & \begin{matrix} a & b & c & d & e & f \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \end{matrix} & \begin{bmatrix} 0 & 1.11 & 2.54 & 3.79 & 5.46 & 6.57 \\ 6.88 & 0 & 1.43 & 2.68 & 4.35 & 5.46 \\ 5.46 & 6.57 & 0 & 1.25 & 2.92 & \mathbf{4.03} \\ 4.21 & 5.32 & 6.75 & 0 & 1.67 & 2.78 \\ 2.54 & 3.65 & 5.08 & 6.33 & 0 & 1.11 \\ 1.43 & 2.54 & \mathbf{3.97} & 5.22 & 6.88 & 0 \end{bmatrix} \end{matrix}.$$

Using the hitting time matrix, we identify the pair of vertices that minimizes the maximal mutual hitting time by computing

$$(5.13) \quad (i^*, j^*) = \arg \min_{i \neq j} \max(H_{i,j}, H_{j,i}), \quad (i^*, j^*) = (c, f)$$

Interpreting the matrix in this way helps identifies the pair of vertices minimizing the maximal mutual hitting time, telling us the two most probable halves that fill the 1 dimensional feature. In this case, it is the simplices $(c \rightarrow f)$ and $(f \rightarrow c)$ that together fill the loop. Furthermore, because the $(f \rightarrow c)$ simplex is more probable than the $(c \rightarrow f)$ simplex, we know that the latter kills the feature. Comparing this prediction with the barcode does show the cycle dies at probability at $0.432 = 0.8 \times 0.6 \times 0.9$. Thus, the hitting time characterization correctly identifies the death simplex associated with the persistence interval.

6. Conclusion. We wanted to use Dowker persistent homology to study a cyclic directed network where edge weights are given as probabilities. In traditional persistent homology, simplices are added by an increasing filtration, whereas for probabilistic edges, it would be desired to have higher probability edges added first. To solve this issue, we used a negative logarithmic transform on the probabilities to turn them into increasing edge weights. This gives an analogous construction where the simplex filtration weight of any simplex can be translated to the product of the probabilities of edges along a path connecting those vertices. We also studied what happens when a single bidirectional or reverse edge is added to the cycle, allowing, at one vertex, flow in both directions. This can create earlier birth and death

times, and in some circumstances, it causes the cycle to be born and die at the same time giving no persistent homology at all.

Once we were able to view the edges probabilistically, it became possible to think about the cycle as a Markov chain. When doing so, we discovered an interesting relationship between the Dowker persistent homology of a consistently oriented cycle and the hitting times of the Markov chain. It is possible to read both the birth time and the death simplex off of the hitting time matrix. The death simplex is the full simplex between the two vertices that have the smallest maximum pairwise hitting time. To test this, we implemented an algorithm to compute hitting times. We hope that this result can inspire future connections between persistent homology theory and the theory of Markov chains.

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REFERENCES

- [1] U. BAUER AND M. LESNICK, *Induced matchings of barcodes and the algebraic stability of persistence*, in Proceedings of the Thirtieth Annual Symposium on Computational Geometry, SoCG '14, ACM, 2014, pp. 355–364, <https://doi.org/10.1145/2582112.2582168>, <https://doi.org/10.1145/2582112.2582168>.
- [2] J.-D. BOISSONNAT AND C. MARIA, *The simplex tree: An efficient data structure for general simplicial complexes*, Algorithmica, 70 (2014), pp. 406–427, <https://doi.org/10.1007/s00453-014-9887-3>.
- [3] G. CARLSSON, *Topology and data*, Bulletin of the American Mathematical Society, 46 (2009), pp. 255–308, <https://doi.org/10.1090/S0273-0979-09-01249-X>.
- [4] J. G. CARLSSON, *Applying topological data analysis to logistics systems analysis*, UC Davis: national Center for Sustainable Transportation, (2024), <https://doi.org/https://doi.org/10.7922/G20C4T4B>, <http://dx.doi.org/10.7922/G20C4T4B>Retrieved from <https://escholarship.org/uc/item/7m0347nd>.
- [5] H.-W. CHANG, S. BACALLADO, V. S. PANDE, AND G. E. CARLSSON, *Persistent topology and metastable state in conformational dynamics*, PLOS ONE, 8 (2013), pp. 1–10, <https://doi.org/10.1371/journal.pone.0058699>, <https://doi.org/10.1371/journal.pone.0058699>.
- [6] S. CHOWDHURY AND F. MÉMOLI, *A functorial dowker theorem and persistent homology of asymmetric networks*, Journal of Applied and Computational Topology, 2 (2018), pp. 115–175, <https://doi.org/10.1007/s41468-018-0020-6>, <https://link.springer.com/article/10.1007/s41468-018-0020-6>.
- [7] C. H. DOWKER, *Homology groups of relations*, Annals of Mathematics, 56 (1952), pp. 84–95, <https://doi.org/10.2307/1969768>, <https://doi.org/10.2307/1969768>.
- [8] H. EDELSBRUNNER AND J. HARER, *Computational Topology: An Introduction*, American Mathematical Society, Providence, RI, 2010.
- [9] H. EDELSBRUNNER, D. LETSCHER, AND A. ZOMORODIAN, *Topological persistence and simplification*, Discrete & Computational Geometry, 28 (2002), pp. 511–533, <https://doi.org/10.1007/s00454-002-2885-2>, <https://api.semanticscholar.org/CorpusID:9191014>.
- [10] D. FARANDA, T. LACOMBE, N. OTTER, AND K. STRØMMEN, *Climate science at the interface between topological data analysis and dynamical systems theory*, Notices of the American Mathematical Society, 71 (2024), pp. 267–271, <https://doi.org/10.1090/noti2864>, <https://www.ams.org/notices/202402/rnoti-p267.pdf>.
- [11] W. K. HASTINGS, *Monte carlo sampling methods using markov chains and their applications*, Biometrika, 57 (1970), pp. 97–109, <https://doi.org/10.1093/biomet/57.1.97>, <https://doi.org/10.1093/biomet/57.1.97>, <https://arxiv.org/abs/https://academic.oup.com/biomet/article-pdf/57/1/97/23940249/57-1-97.pdf>.
- [12] A. HATCHER, *Algebraic Topology*, Cambridge University Press, Cambridge, 2002.
- [13] B. JONES, C. POTVIN, S. TANWEER, K. KURIANSKI, AND T. TIMOFEYEV, *Cactis: Coarse-grained asymmetric cycles in time series*. <https://github.com/cactismath/CACTIS>, 2025.
- [14] B. P. KITCHENS, *Symbolic Dynamics: One-sided, Two-sided and Countable State Markov Shifts*, Univer-

- sitext, Springer Berlin, Heidelberg, 1 ed., 1998, <https://doi.org/10.1007/978-3-642-58822-8>. eBook ISBN: 978-3-642-58822-8.
- [15] M. Q. LE AND D. TAYLOR, *Persistent homology of convection cycles in network flows*, Phys. Rev. E, 105 (2022), p. 044311, <https://doi.org/10.1103/PhysRevE.105.044311>, <https://link.aps.org/doi/10.1103/PhysRevE.105.044311>.
- [16] D. A. LEVIN, Y. PERES, AND E. L. WILMER, *Markov Chains and Mixing Times*, American Mathematical Society, Providence, RI, 2 ed., 2009.
- [17] S. MALETIĆ, Y. ZHAO, AND M. RAJKOVIĆ, *Persistent topology of complex networks*, Journal of Statistical Mechanics: Theory and Experiment, (2016), p. 053304, <https://doi.org/10.1088/1742-5468/2016/05/053304>.
- [18] D. E. MUÑOZ, E. MUNCH, AND F. A. KHASAWNEH, *The walk-length filtration for persistent homology on weighted directed graphs*, 2026, <https://arxiv.org/abs/2506.22263>, <https://arxiv.org/abs/2506.22263>.
- [19] C. E. SHANNON, *A mathematical theory of communication*, The Bell System Technical Journal, 27 (1948), pp. 379–423, <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>.
- [20] K. STROMMEN AND T. N. PALMER, *Signal and noise in regime systems: A hypothesis on the predictability of the north atlantic oscillation*, Quarterly Journal of the Royal Meteorological Society, 145 (2019), pp. 147–163, <https://doi.org/https://doi.org/10.1002/qj.3414>, <https://rmets.onlinelibrary.wiley.com/doi/abs/10.1002/qj.3414>, <https://arxiv.org/abs/https://rmets.onlinelibrary.wiley.com/doi/pdf/10.1002/qj.3414>.
- [21] T. TIMOFEYEV, C. POTVIN, B. JONES, K. M. KURIANSKI, M. LOPEZ, AND S. TANWEER, *Asymmetrically weighted dowker persistence and applications in dynamical systems*, arXiv preprint, (2025).